RADHIKA AMAR DESAI

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Education

Degree/Certificate	Institute/Board	CGPA/Percentage	Year
B.Tech. (CSE)	VIT, Chennai	9.24/10	2022-2026
Senior Secondary Certificate	CBSE Board	97.16%	2022

Work Experience

- **Airbus** *Jan'26 - Present*
  Computer Vision Intern Bangalore, Karnataka
  – Conducting research on coreset selection for efficient training of deep learning models, focusing on data-efficient learning and scalability
  – Benchmarking existing coreset selection and data pruning techniques across standard datasets to evaluate performance, generalization, and computational trade-offs
  – Proposed a novel coreset selection method based on embedding-space analysis, currently under evaluation on benchmark datasets
- **Stryker Global Technology Center R&D** *Feb'25 - Jan'26*
  Research Associate Bangalore, Karnataka (Hybrid mode)
  – Contributing to a Proof of Concept (POC) in the medical imaging domain by integrating computer vision and deep learning approaches for clinical workflows.
  – Designed and implemented deep learning pipelines in PyTorch for tasks including image segmentation, image registration, temporal consistency, and inpainting, ensuring robustness and adherence to clinical data standards.
  – Built automated data preparation pipelines for large-scale medical imaging datasets, including preprocessing, augmentation, and annotation handling, leveraging Azure ML for reproducibility and scalability
  – Conducted model training, hyperparameter tuning, and experimentation on Azure ML, tracking performance metrics and optimizing for accuracy, efficiency, and deployment-readiness.
  – Collaborating with cross-functional teams to document findings and present insights for future clinical validation and productization.
  – Participating in labeling medical imaging datasets while coordinating and reviewing data annotation work, ensuring quality and consistency; provided guidance and feedback to fellow interns to maintain high annotation standards.
- **Schneider Electric** *May'24 - Jul'24*
  Project Trainee Doha, Qatar
  – Automated data preprocessing workflows for industrial control system logs (Foxboro controllers), improving efficiency and reducing manual effort.
  – Developed automation tools to test communication between Triconex and Foxboro systems, enhancing reliability in Emergency Shutdown Systems (ESD).
  – Created multiple Visual Basic automations for repetitive Excel tasks related to supply chain and internal reporting, improving efficiency for project teams.
  – Assisted in Factory Acceptance Testing (FAT) for large-scale industrial systems, gaining experience with audit-like validation processes.

Research Projects

- **Discriminative Span** *Jan'25-Present*
  Research paper under preparation for submission to top-tier conferences on synthetic data
  – Developing a novel framework to analyze difference vectors in latent embedding spaces, quantifying the impact of synthetic data on downstream model performance.
  – Designing and implementing a modular pipeline integrating feature extraction (ResNet/EfficientNet/MobileNet), statistical analysis, and supervised classification, with ongoing ablation studies and extensions to domain adaptation and data selection.
  – Conducting extensive experiments on benchmark datasets (e.g., CycleGAN), evaluating alignment quality, classifier separability, and generalization, achieving up to ∼96% test accuracy under high-alignment conditions
  – Paper: Discriminative Span as a Predictor of Synthetic Data Utility via Classifier Reconstruction
- **Synthetic Data vs Generalization (CXR Study)** *Jan'25-Present*
  Research paper under preparation for submission to top-tier conferences on synthetic data
  – Investigated the limitations of synthetic data in medical imaging, showing that models trained solely on synthetic CXR data fail to generalize despite strong performance on standard distributional metrics.
  – Developed a pseudo-paired synthetic data generation pipeline leveraging text-conditioned image editing for controlled pathology injection.
  – Developed a task-aware evaluation metric to measure how well synthetic data represents real data by modifying the k-center greedy objective and incorporating uncertainty-based weighting.

- **Detection of Fraudulent Transactions In Ethereum** *Oct'24-Apr'25*
  Research paper accepted into the International Conference On AI-Powered Security For Metaverse.
  – Developed an anomaly detection pipeline for fraudulent financial transactions using Python and deep learning, achieving 99.9% accuracy.
  – Designed data preprocessing and feature engineering workflows for large-scale financial datasets.
  – Repository: Ethereum_Fraud_Transaction_Detection

- **Automatic PCOS Detection System using Ultrasound Images of Ovaries** *Jun'24-Jul'25*
  Research paper accepted into the International Conference On Communication And Smart Devices
  – Built a data preprocessing and anomaly detection pipeline for ultrasound datasets using Python and OpenCV.
  – Developed a novel segmentation technique to standardize noisy medical images.
  – Applied machine learning models (scikit-learn, ML libraries) achieving 99% accuracy in classification.
  – Paper: Automatic_PCOS_Detection_Using_Ultrasound_images.git

Technical Skills

- **Programming & Databases**: Python, SQL, C, C++, Visual Basic | **Data Analytics & Visualization**: Power BI, Excel, OpenCV | **Embedded**: Arduino | **Tools**: TriStation, FoxView | **Libraries**: PyTorch, OpenCV | **Cloud**: AWS, Azure ML | **Machine Learning & Others**: Data Preprocessing, Anomaly Detection, Image Segmentation, Automation Workflows

Referrals

- **VIT Research Faculties:** chanthini.baskar@vit.ac.in, modigari.narendra@vit.ac.in
- **Schneider Electric Project Manager:** ritwik.bhattacharjee@se.com
- **Stryker Senior Manager:** venkatachalam.vr@stryker.com